Track A: Cardiovascular Risk Visualization and Prediction

Project documentation: dataset preprocessing, model architecture, 3D pipeline, usage and evaluation

Clinical safety notice. Predictions are for decision support and educational purposes only. They are not a substitute for formal diagnostic imaging or clinical judgement.

1. Overview

The system predicts overall coronary artery disease (CAD) and stenosis of the three main coronary arteries (LAD, LCX, RCA) from demographic, examination, ECG, laboratory and echocardiographic data, and maps the predictions onto an interactive 3D heart built from real BodyParts3D anatomy. Four independent classifiers feed a Flask API; a clinical dashboard shows the probabilities, an exact feature level explanation and the patient's physiological measurements next to the 3D view. Clicking a vessel in 3D switches the explanation to that vessel, and every new prediction recolours the arteries immediately.

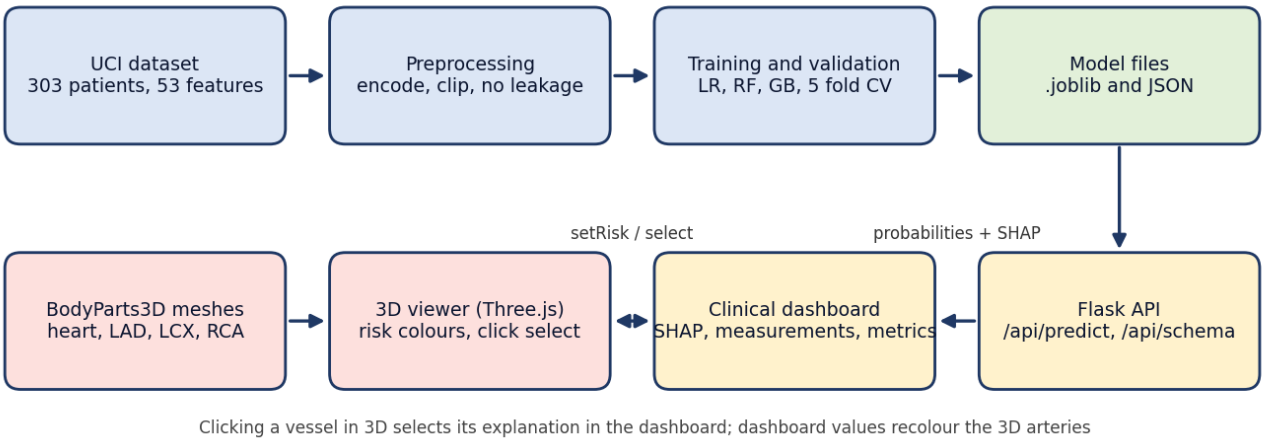


Figure 1. System architecture. Model files are produced once by the training pipeline and served by the API; the viewer also runs standalone with the same model embedded.

2. Dataset and preprocessing

Data. The Extension of the Z-Alizadeh Sani dataset (UCI Machine Learning Repository) has 303 patients (176 male, 127 female, age 30 to 86, mean 58.9). Four binary targets are derived from the angiography columns: **CAD** (Cath = CAD, 216 patients, 71%), **LAD** stenotic (177, 58%), **LCX** stenotic (119, 39%) and **RCA** stenotic (114, 38%).

Preprocessing steps (implemented in *train_pipeline.py*):

- **Leakage control.** LAD, LCX, RCA and Cath are removed from the inputs before any model sees the data; an assertion in the script enforces it.
- **Encoding.** Sex becomes 1 for male; Y/N columns become 1/0; valvular disease (VHD) is ordinal (none, mild, moderate, severe); Region with RWMA stays numeric (0 to 4). No values were missing.
- **Redundancy.** Weight and Length are dropped because BMI already contains them; the app asks for BMI.
- **Outliers.** Every feature is clipped to its 1st and 99th percentile, and the same bounds are applied at prediction time. Values outside the training range trigger a warning in the API.
- **Result.** 53 input features grouped as demographics, history and risk factors, physical examination, symptoms, ECG, laboratory and echocardiography. Region with RWMA is used only as a clinical feature; it is not an anatomical coordinate, so the 3D map is driven by the model outputs, not by RWMA.

3. Model architecture and validation

Architecture. One binary classifier per target (CAD, LAD, LCX, RCA): feature standardisation followed by class balanced logistic regression. The penalty (L1 or L2) and strength were chosen per target by 5 fold stratified cross validation on the training split only (CAD: L1, C 0.3; LAD: L1, C 0.1; LCX: L2, C 0.01; RCA: L1, C 0.1). Random forest (400 trees) and gradient boosting (150 shallow trees) were trained as comparison models.

Validation. Stratified 80/20 train/test split per target (seed 42, 242 training and 61 test patients). All tuning and cross validation used the training part only; the test split was scored once. The deployed weights are then refit on all 303 patients, so the hold out numbers below describe the same model recipe trained on 242 patients. Metrics are accuracy, precision, recall, F1 and ROC-AUC at a 0.5 threshold, with bootstrap 95% confidence intervals (2000 resamples) for AUC.

Why logistic regression is deployed. In cross validation its AUC is within about 0.03 of the tree models on every target (and higher on RCA), it exports exactly to JSON so the browser can run the same model, and its explanations are exact: for a linear model the SHAP value of a feature equals its coefficient times its standardised value, and the contributions plus the baseline sum exactly to the model's log odds (checked by an automated test).

4. Evaluation results

Table 1. Hold out performance of the deployed logistic regression (last column: mean 5 fold CV AUC on the training split for all three model families).

Target	Test n	Pos. rate	AUC	AUC 95% CI	Acc.	Prec.	Recall	F1	CV AUC (LR / RF / GB)
CAD	61	0.71	0.897	0.80 to 0.97	0.820	0.900	0.837	0.867	0.929 / 0.932 / 0.929
LAD	61	0.58	0.854	0.75 to 0.94	0.705	0.821	0.639	0.719	0.833 / 0.837 / 0.821
LCX	61	0.39	0.660	0.52 to 0.79	0.607	0.500	0.625	0.556	0.714 / 0.717 / 0.678
RCA	61	0.38	0.788	0.66 to 0.90	0.705	0.586	0.739	0.654	0.740 / 0.717 / 0.679

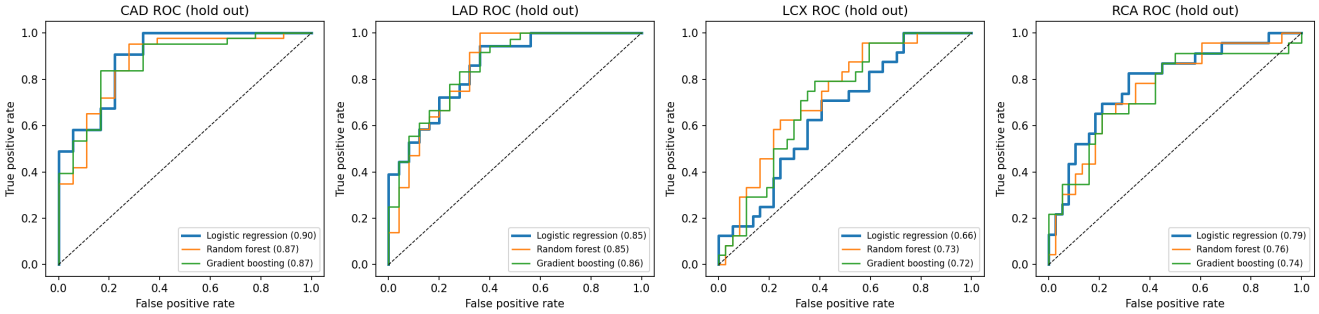


Figure 2. Hold out ROC curves for the three model families on each target (deployed model in bold).

Reading the results. Overall CAD is predicted well (AUC 0.90) and LAD reasonably (0.85). RCA is moderate (0.79). **LCX is weak** (hold out AUC 0.66, interval 0.52 to 0.79; cross validation 0.71) and no model family does clearly better in cross validation, so the data, not the algorithm, limits it. With only 61 test patients the intervals are wide, so cross validation numbers are the more stable guide. Because class weights are balanced, outputs are risk scores that rank patients, not calibrated probabilities (see *plots/calibration.png*); the dashboard bands (below 33% low, 33 to 66% moderate, above 66% high) are presentation bands, not clinical thresholds.

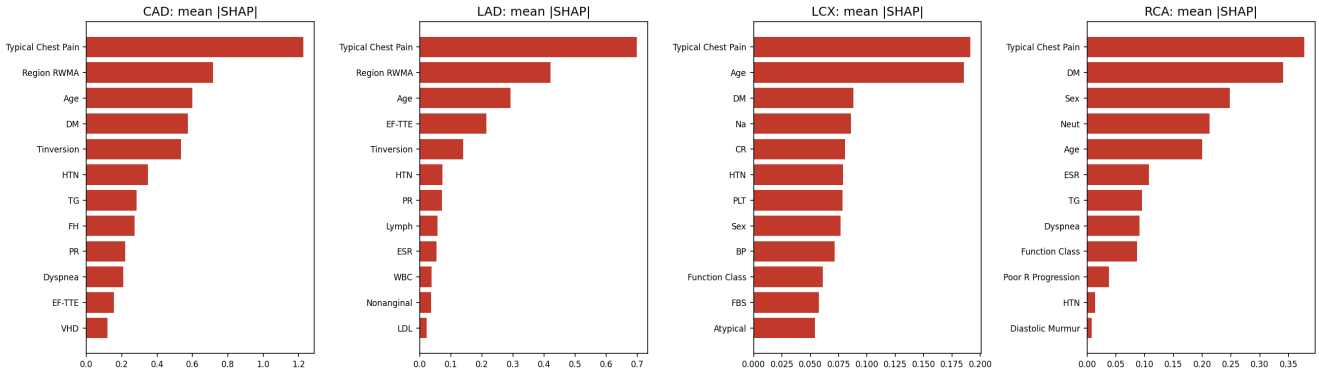


Figure 3. Overall feature importance (mean absolute SHAP value over the 303 patients) for each target.

Typical chest pain is the strongest driver for all four targets. Region with wall motion abnormality, age, diabetes and T inversion follow for overall CAD; for LAD, wall motion abnormality, age and ejection fraction matter most; for LCX and RCA no single feature dominates, which matches their weaker performance.

5. 3D visualization pipeline

Anatomy. Real open source BodyParts3D meshes (Database Center for Life Science) are used: the heart from three parts (FJ2428, FJ2438, FJ2439), the LAD (FJ2631), the LCX from six pieces (FJ2649 to FJ2654) and the RCA (FJ2723). A manifest assigns each file to a role; several files per role are combined and coloured together.

Automatic orientation. The meshes arrive in the source coordinate system (millimetres, Z up). The viewer derives the patient frame from the anatomy itself: patient left is the direction from the RCA toward the LAD and LCX, anterior is from the heart centre toward the LAD, and up is the source Z axis. It then centres the heart and scales it to a fixed size. This was checked on the real files with front, side and top projections, and on synthetic poses including upside down meshes.

Spatial risk mapping. The predicted probability of each vessel sets its colour (green to red) and a glow that grows with risk; the heart body is tinted by overall CAD risk. Because the dataset has no lesion location, the real vessel meshes are not narrowed at a point; risk is shown by colour and glow only.

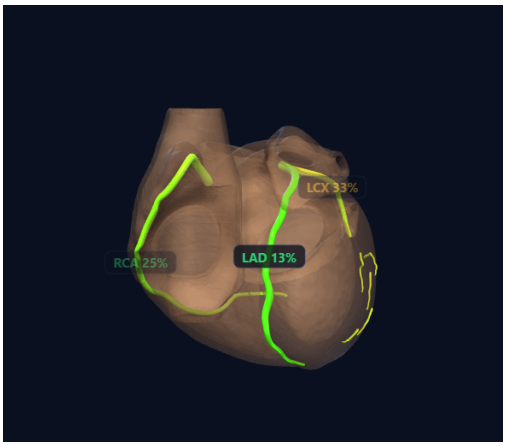


Figure 4. Viewer with real BodyParts3D heart, RCA, LAD and LCX coloured by predicted risk.

- **Interaction.** Drag to rotate, wheel or pinch to zoom, click a vessel or the heart to select it, or use the Focus buttons. Vessel labels show the live percentage and fade when the vessel is behind the heart.
- **No dedicated GPU.** The scene uses Three.js (WebGL) with about 70 thousand triangles, a standard material and no post processing; orbit controls are custom, so there are no extra dependencies beyond one script.
- **Fallback and extensibility.** If no mesh files are present a built in schematic heart is shown. New anatomical structures or other mesh sets are added by listing files in *static/models/bp3d/manifest.json*; the vessel to model output mapping does not change.
- **Attribution.** The mesh files state the license Creative Commons Attribution-Share Alike 2.1 Japan: "BodyParts3D, (c) The Database Center for Life Science licensed under CC Attribution-Share Alike 2.1 Japan". The viewer and dashboard display this line, and modified copies of the meshes must carry the same license.

6. Backend, clinical dashboard and integration

The Flask backend (*app.py*, *predictor.py*) loads the trained models once and also serves the dashboard and the viewer from the same origin.

Endpoint	Purpose
GET /api/schema	The 53 features grouped for forms with type, range, default, unit and typical range
POST /api/predict	Features in; probabilities, risk band and SHAP contributions for CAD, LAD, LCX, RCA out. Missing values use the training median; invalid input returns HTTP 400
GET /api/metrics	Hold out and cross validation metrics, global feature importance
GET /api/samples, /api/health	30 real dataset patients with true labels for demos; liveness check

Dashboard panels. (1) Patient form with all 53 features, a picker for real dataset patients, a Predict button (it asks you to choose a patient first) and a Reset button. (2) The embedded 3D viewer, with grey arteries until the first prediction. (3) Predicted risk for overall CAD and each vessel with a band. (4) The ten strongest contributions for the selected region as a diverging bar chart (red raises risk, green lowers it). (5) A table of physiological measurements (age, BMI, blood pressure, pulse, glucose, lipids, ejection fraction, kidney and blood counts) with value, typical adult range, status, signed contribution and share of the total contribution. (6) Hold out metrics and global feature importance. The safety notice is always visible.

Model to viewer integration. The dashboard sends the form to */api/predict*, then calls *CADViewer.setRisk(probabilities)* on the embedded viewer, *setPulse* for the heartbeat rate and *select* for the highlighted vessel. Clicking a vessel fires a *cad:select* event that the dashboard uses to switch its explanation. The same *CADViewer* API lets any other model drive the 3D view.

7. Usage instructions

Install and run. The app needs Python 3.10 or newer and only Flask, NumPy and pandas (*requirements.txt*); the trained models are shipped as plain numbers, so scikit-learn is needed only to retrain (*requirements-train.txt*, written with Python 3.12 and scikit-learn 1.8):

```
cd cad_project; pip install -r requirements.txt
python test_api.py       # backend self test
python app.py            # open http://127.0.0.1:8000

# optional, to retrain: pip install -r requirements-train.txt
python train_pipeline.py   # train, validate, export model files and metrics.json
python evaluate_plots.py  # ROC, confusion matrices, calibration, importance, results table
```

- **Standalone viewer.** Open *cad_3d_viewer_standalone.html* in a modern browser (meshes and model are embedded; one script is loaded from the internet).
- **Workflow.** Pick a dataset patient (or edit features), then press Predict; Reset empties the form and all result panels, and Predict then asks for a patient again; the four probabilities and the 3D colours update; click LAD, LCX, RCA or the heart to see why; the measurements table and metrics are in the bottom panel. The side panels and the bottom panel can be resized by dragging their splitters.
- **Reproducibility.** All randomness uses seed 42; rerunning the pipeline reproduced identical metrics.

8. Limitations and responsible use

- Single centre data of 303 patients with suspected CAD; results may not transfer to other populations.
- LCX performance is weak, test sets are small and scores are not calibrated probabilities.
- The dataset has no lesion location or per vessel severity, so the 3D map shows model risk per vessel, not a measured lesion. The anatomy is a generic model, not patient specific.
- The tool is for education and decision support only and must not replace formal diagnostic imaging or clinical judgement.

References

Z-Alizadeh Sani dataset extension, UCI Machine Learning Repository. BodyParts3D, The Database Center for Life Science, https://dbarchive.biosciencedbc.jp/en/bodyparts3d/. Three.js r128. scikit-learn 1.8.